Brain and Cognitive Science Club

Summer Project 2025

Soch Vichar Sangh
SVS

Project Idea

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1 Introduction

Understanding how the human brain processes naturalistic audiovisual experiences is a fundamental challenge in computational neuroscience. This project presents a TRIBE-inspired multimodal brain encoding framework for the Algonauts 2025 Challenge, aiming to predict parcel-wise fMRI responses recorded while participants watched episodes of Friends.

The framework combines textual and visual information by extracting semantic embeddings using pretrained Sentence-BERT and spatiotemporal video features using a pretrained VideoMAE model. These features are temporally aligned with preprocessed fMRI signals, fused into a shared representation, and processed by a Transformer Encoder to model temporal relationships between stimuli and brain activity.

To improve generalization, the model employs modality dropout, AdamW optimization, learning rate scheduling, early stopping, and Stochastic Weight Averaging (SWA). Final predictions are obtained using a softmax-weighted ensemble of independently trained models. The proposed approach demonstrates the effectiveness of integrating language and vision representations for predicting neural responses to complex audiovisual stimuli.

2 Overall Architecture

The proposed model consists of six major stages, beginning with multimodal stimulus processing and ending with parcel-wise fMRI prediction.

1. Data preprocessing and temporal alignment
2. Text feature extraction using Sentence-BERT
3. Video feature extraction using VideoMAE
4. Multimodal feature fusion
5. Temporal modeling using a Transformer Encoder
6. Linear prediction of parcel-wise fMRI responses

The textual transcripts corresponding to each movie segment are converted into dense semantic embeddings using the pretrained Sentence-BERT model (`all-MiniLM-L6-v2`). Simultaneously, video clips are processed by a pretrained VideoMAE model to extract high-level spatiotemporal representations.

Both modalities are projected into a common feature space and concatenated to form a multimodal representation. These fused embeddings are then passed through multiple Transformer Encoder layers that learn temporal relationships across consecutive stimulus segments.

Finally, the output representations are mapped to parcel-wise brain responses using a fully connected prediction head. The complete architecture is trained end-to-end using multimodal supervision from preprocessed fMRI recordings.

3 Technology Stack

The proposed framework is implemented using Python 3.10 and the PyTorch deep learning framework. Numerical computations and data manipulation are performed using NumPy and Pandas, while Matplotlib is employed for visualization.

Textual representations are extracted using the pretrained Sentence-BERT model (`all-MiniLM-L6-v2`), and visual features are obtained using the pretrained VideoMAE model. Optimization is carried out using the AdamW optimizer with cosine annealing learning rate scheduling. Model evaluation is based on the Pearson correlation coefficient between predicted and observed parcel-wise fMRI responses.

4 Feature Extraction

The proposed framework extracts complementary semantic and visual representations from textual transcripts and video clips. Each modality is processed independently using pretrained foundation models before multimodal fusion.

4.1 Text Features (Sentence-BERT)

Textual transcripts corresponding to each movie segment are encoded using the pretrained Sentence-BERT model (`all-MiniLM-L6-v2`). Sentence-BERT generates dense contextual embeddings that capture semantic relationships between words and sentences. These fixed-length embeddings serve as high-level language representations for the subsequent multimodal fusion stage.

4.2 Video Features (VideoMAE)

Visual representations are extracted from video clips using the pretrained VideoMAE model. VideoMAE is a masked autoencoder designed for learning spatiotemporal features from video sequences through self-supervised pretraining. The extracted embeddings capture motion, object appearance, scene context, and temporal dynamics, providing rich visual information complementary to the textual representations.

5 Multimodal Fusion Network

The text and video embeddings are first projected into a shared feature space using linear projection layers. The projected features are concatenated to form a unified multimodal representation that combines semantic and visual information.

The fused representation is processed by multiple Transformer Encoder layers, allowing the model to capture long-range temporal dependencies across consecutive stimulus segments. Multi-head self-attention enables effective interaction between language and vision features, resulting in richer representations for predicting neural activity.

6 Training Strategy

The multimodal Transformer network is trained to predict parcel-wise fMRI responses from synchronized audiovisual features. Mean Squared Error (MSE) is used as the training objective, while Pearson correlation on the validation set is employed to monitor prediction quality. Training is performed for multiple epochs using mini-batch optimization, with several regularization techniques incorporated to improve generalization.

6.1 Modality Dropout

To improve robustness, modality dropout is applied during training by randomly masking either the text features or the video features with predefined probabilities. This prevents the network from relying excessively on a single modality and encourages learning complementary information from both language and vision. Consequently, the model remains effective even when one modality provides limited information.

6.2 Optimization

Model parameters are optimized using the AdamW optimizer, which combines adaptive learning rates with decoupled weight decay for improved generalization. A cosine annealing learning rate scheduler gradually reduces the learning rate during training, enabling stable convergence and preventing oscillations in the optimization process.

6.3 Stochastic Weight Averaging

Stochastic Weight Averaging (SWA) is employed during the final stages of training to improve generalization. Instead of using the final model checkpoint alone, SWA averages parameters from multiple checkpoints obtained near convergence. This produces flatter optimization minima and generally yields better validation performance than conventional optimization.

6.4 Early Stopping

Early stopping is used to prevent overfitting by monitoring the validation Pearson correlation during training. If the validation performance does not improve for a predefined number of epochs, training is terminated and the best-performing model is retained for evaluation.

7 Ensemble Learning

To further improve prediction accuracy, multiple independent models are trained using different random initializations and data shuffling configurations. Each model generates parcel-wise predictions, which are combined using a softmax-weighted ensemble based on validation performance. The ensemble reduces prediction variance and consistently outperforms individual models by leveraging the complementary strengths of multiple learned representations.

8 Conclusion

This work presents a TRIBE-inspired multimodal brain encoding framework for predicting neural responses to naturalistic audiovisual stimuli. By combining semantic representations from Sentence-BERT with spatiotemporal visual features extracted using VideoMAE, the proposed approach effectively models complex brain activity through a Transformer-based fusion network. Training strategies such as modality dropout, AdamW optimization, SWA, and ensemble learning further improve generalization and prediction accuracy. The framework demonstrates the potential of multimodal deep learning for advancing computational neuroscience and understanding brain responses to real-world experiences

9 Our Work

As an initial step toward building a multimodal brain encoding framework, we developed a text-only baseline using the Algonauts 2025 dataset. The dataset contains synchronized movie stimuli, textual transcripts, and parcel-wise fMRI recordings collected while participants watched episodes of the television series *Friends*. In this phase, only the transcript modality was utilized, providing a benchmark before incorporating visual information.

Each transcript was processed using the pre-trained Sentence-BERT (`all-MiniLM-L6-v2`) model to generate 384-dimensional contextual embeddings. Consecutive transcript segments were grouped into sequences of length eight, enabling the model to capture temporal context within the dialogue. These embeddings were temporally aligned with the corresponding preprocessed fMRI recordings, where each sample consisted of responses from 1000 cortical parcels.

The complete dataset contained 404,104 training samples, which were divided into 323,283 training samples and 80,821 validation samples using an 80:20 split. The neural encoding model was trained for 15 epochs using the AdamW optimizer with automatic mixed-precision training to improve computational efficiency. Stochastic Weight Averaging (SWA) was applied during the final stage of training to improve model generalization and stability.

Model performance was evaluated using the Pearson correlation coefficient between the predicted and observed parcel-wise fMRI responses. Using only textual transcript features, the proposed baseline achieved a final validation Pearson correlation of **0.10047**. Although the model relied solely on semantic information extracted from dialogue, it successfully captured meaningful neural representations and established a strong baseline for subsequent multimodal experiments.

The next phase of this work extends the framework by integrating VideoMAE-based visual embeddings with Sentence-BERT representations through a multimodal Transformer architecture. The combination of language and visual features is expected to better model the complex audiovisual stimuli presented during movie viewing and improve brain response prediction accuracy.